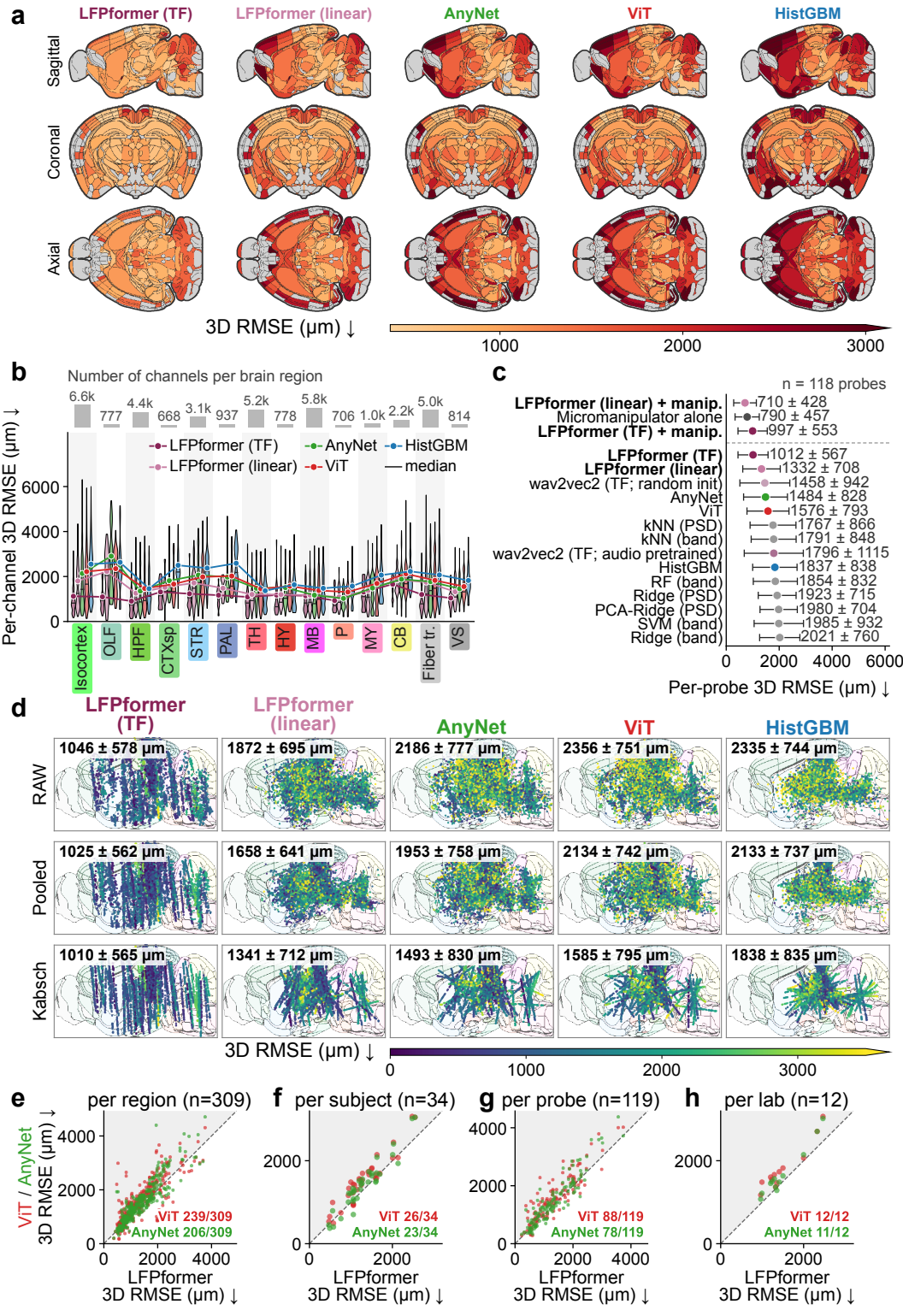

a
LFPformer (TF)
LFPformer (linear)
AnyNet
ViT
HistGBM
Sagittal
Coronal
Axial
3D RMSE (µm) ↓
1000
2000
3000
b
Number of channels per brain region
6.6k
777
4.4k
668
3.1k
937
5.2k
778
5.8k
706
1.0k
2.2k
5.0k
814
LFPformer (TF)
LFPformer (linear)
AnyNet
ViT
HistGBM
median
Per-channel 3D RMSE (µm) ↓
6000
4000
2000
0
Isocortex
OLF
HPF
CTXsp
STR
PAL
TH
HY
MB
P
MY
CB
Fiber tr.
VS
c
n = 118 probes
LFPformer (linear) + manip. 710 ± 428
Micromanipulator alone 790 ± 457
LFPformer (TF) + manip. 997 ± 553
LFPformer (TF) 1012 ± 567
LFPformer (linear) 1332 ± 708
wav2vec2 (TF; random init) 1458 ± 942
AnyNet 1484 ± 828
ViT 1576 ± 793
kNN (PSD) 1767 ± 866
kNN (band) 1791 ± 848
wav2vec2 (TF; audio pretrained) 1796 ± 1115
HistGBM 1837 ± 838
RF (band) 1854 ± 832
Ridge (PSD) 1923 ± 715
PCA-Ridge (PSD) 1980 ± 704
SVM (band) 1985 ± 932
Ridge (band) 2021 ± 760
0
2000
4000
6000
Per-probe 3D RMSE (µm) ↓
d
LFPformer (TF)
LFPformer (linear)
AnyNet
ViT
HistGBM
RAW
Pooled
Kabsch
1046 ± 578 µm
1872 ± 695 µm
2186 ± 777 µm
2356 ± 751 µm
2335 ± 744 µm
1025 ± 562 µm
1658 ± 641 µm
1953 ± 758 µm
2134 ± 742 µm
2133 ± 737 µm
1010 ± 565 µm
1341 ± 712 µm
1493 ± 830 µm
1585 ± 795 µm
1838 ± 835 µm
3D RMSE (µm) ↓
0
1000
2000
3000
e
per region (n=309)
ViT / AnyNet 3D RMSE (µm) ↓
4000
2000
0
ViT 239/309
AnyNet 206/309
0
2000
4000
LFPformer 3D RMSE (µm) ↓
f
per subject (n=34)
2000
0
ViT 26/34
AnyNet 23/34
0
2000
LFPformer 3D RMSE (µm) ↓
g
per probe (n=119)
4000
2000
0
ViT 88/119
AnyNet 78/119
0
2000
4000
LFPformer 3D RMSE (µm) ↓
h
per lab (n=12)
2000
0
ViT 12/12
AnyNet 11/12
0
2000
LFPformer 3D RMSE (µm) ↓